Question # 8 of 10 (Start time: 03:21:51 PM, 05 July 2024)

Total Marks: 1

Total derivative of the function

$$y = f(g(w), w)$$

is:

Select the correct option

Reload Math Equations

- $\frac{dy}{dw} = \frac{dx}{dw} + \frac{dw}{dw}$
- $\frac{dy}{dw} = dx + f_w \cdot \frac{dw}{dw}$
- $\frac{dy}{dw} = f_x \cdot \frac{dx}{dw} + dx$
- $\frac{dy}{dw} = f_x \cdot \frac{dx}{dw} + f_w \cdot \frac{dw}{dw}$

Click to Save Answer & Move to Next Question

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85%

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Question # 6 of 10 (Start time: 03:52:00 PM, 05 July 2024)

One of the major components of money supply is:

Select the correct option

- Discount rate
- Bank deposits
- Bank deposits
- Bank deposits

Click to Save An

sec(s)

ECO606 - Mathematical Economics I (Quiz No 3) **Quiz Start Time: 03:15 PM, 05 July 2024**

Question # 6 of 10 (Start time: 03:19:26 PM, 05 July 2024) **Total Marks: 1**

What happens as a result of technological changes in the production process?

Select the correct option

- Production function shifts inward
- Production function shifts outward
- No change in the production function
- Production function remain at its initial condition

Click to Save Answer & Move to Next Question

ECO606 - Mathematical Economics I (Quiz No 3)

Quiz Start Time: 03:15 PM, 05 July 2024

Question # 3 of 10 (Start time: 03:17:01 PM, 05 July 2024)

Total Marks: 1

Formula to calculate average cost is:

Select the correct option

- Total cost * Output (Q)
- Marginal cost * Output (Q)
- Marginal cost / Output (Q)
- Total cost / Output (Q)

Question # 4 of 10 (Start time: 03:17:55 PM, 05 July 2024) **Total Marks:** 1

Which of the following is used to relate supply of goods with producer's profit?

Select the correct option

- Hotelling's Lemma
- Shephard Lemma
- Roy's identity
- Envelope theorem

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 03:18:39 PM, 05 July 2024) **Total Marks:** 1

$$f'(x)$$

shows the:

Select the correct option

Reload Math Equations

- Third derivative of the function $y = f(x)$
- Second derivative of the function $y = f(x)$
- First derivative of the function $y = f(x)$
- Fourth derivative of the function $y = f(x)$

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 03:27:01 PM, 05 July 2024)

In the consumption function; C = 1250 + 0.85Y, marginal propensity to consume is:

Select the correct option

- 1250.85
- 1249.5
- 0.85
- 1250



Type here to search

Question # 4 of 10 (Start time: 03:29:08 PM, 05 July 2024)

Cost of raw material is an example of:

Select the correct option

- Sunk cost
- Fixed cost
- Variable cost
- Opportunity cost



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Time Left 67 sec(s)

ECO606 - Mathematical Economics I (Quiz No 3)

Quiz Start Time: 03:15 PM, 05 July 2024

Question # 1 of 10 (Start time: 03:15:17 PM, 05 July 2024) **Total Marks: 1**

Which of the following is the derivative of total revenue?

Select the correct option

- Fixed revenue
- High revenue
- Marginal revenue
- Average revenue

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 03:23:08 PM, 05 July 2024)

Which of the following shows the differential of

$$y = f(x) = Ax^{\alpha} + \beta$$

?

Select the correct option

○	$dy = (\alpha + \beta)Ax^{\alpha-1}.dx$
○	$dy = \alpha Ax^{\alpha-1}.dx$ ✓
○	$dy = \beta Ax^{\beta-1}.dx$
○	$dx = \alpha Ax^{\alpha-1}.dy$

Click to Save Answ

Question # 10 of 10 (Start time: 03:23:56 PM, 05 July 2024)

Cost which does not change with the level of output is called:

Select the correct option

- Fixed cost
- Opportunity cost
- Variable cost
- Sunk cost

Click to S

stion # 3 of 10 (Start time: 03:27:56 PM, 05 July 2024)

iven the Cobb-Douglas production function;

$$Q = AK^{\alpha}L^{\beta}$$

represents:

ct the correct option

- Output
- Labo
- Capital
- Constant

Q
L
K
α, β

Type here to search

Question # 8 of 10 (Start time: 03:54:18 PM, 05 July 2024)

$$\left(\frac{\partial y}{\partial x}\right)$$

is used to denote:

Select the correct option

- Limit
- Partial derivative
- Simple derivative
- Matrix

Click to Save A

Question # 7 of 10 (Start time: 03:32:17 PM, 05 July 2024)

Formula to calculate total cost is:

Select the correct option

- Average cost × Output (Q)
- Marginal cost + Output (Q)
- Average cost / Output (Q)
- Marginal cost / Output (Q)

Type here to search

Which of the following is necessary for profit maximization level of output?

Select the correct option

- First order derivative of the profit function should be equal to zero
- Third order derivative of the profit function should be equal to zero
- Second order derivative of the profit function should be equal to zero ✓
- Fourth order derivative of the profit function should be equal to zero

Type here to search

ECO606 - Mathematical Economics I (Quiz No 3)

Question # 5 of 10 (Start time: 03:51:24 PM, 05 July 2024)

Given the Cobb-Douglas production function;

$$Q = AK^{\alpha}L^{\beta}$$

L represents:

Select the correct option

○	Output
○	Constant
○	Capital
○	Labor

Click to

Question # 3 of 10 (Start time: 03:49:54 PM, 05 July 2024)

$$y = f(g(x))$$

$$\frac{dy}{dx} = \frac{d}{dx}\{f(g(x))\} = f'(g(x)).g'(x)$$

Given the above expression;

$$g'(x)$$

represents:

Select the correct option

- ○ Outside function
- ○ Derivative of outside function
- ○ Derivative of inside function ✓
- ○ Inside function

Click to Save Answer &

Question # 4 of 10 (Start time: 03:50:36 PM, 05 July 2024)

Which the following represents chain rule of differentiation?

Select the correct option

- ○ $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$ ✓
- ○ $\frac{dy}{dx} = \frac{dy}{du} - \frac{du}{dx}$
- ○ $\frac{dy}{dx} = \frac{dy}{du} / \frac{du}{dx}$
- ○ $\frac{dy}{dx} = \frac{dy}{du} + \frac{du}{dx}$

Click to Save

Question # 2 of 10 (Start time: 03:48:50 PM, 05 July 2024)

High powered money supply is equal to:

Select the correct option

- Bank reserves – Reserve ratio
- Cash holdings – Bank deposits
- Cash holdings + Bank reserves
- Cash holdings + Interest rate



Click to Save Answer & M

Question # 1 of 10 (Start time: 03:47:55 PM, 05 July 2024)

$$\pi = PQ - wK$$

is the profit function of the labor market; what does K show in this function?

Select the correct option

Re

- Price of capital input
- Quantity of capital input
- Quantity of output
- Price of output



Click to Save Answer & Mo

ECO606 – Mathematical Economics I (Quiz No 3)

Question # 1 of 10 (Start time: 03:34:58 PM, 05 July 2024)

$F(Q, K, L) =$

shows:

Select the correct option

- Quadratic function
- Cubic function
- Implicit function ✓
- Cost function

Question # 10 of 10 (Start time: 03:56:07 PM, 05 July 2024)

Formula to calculate average cost is:

Select the correct option

- ○ Marginal cost / Output (Q)
- ○ Total cost / Output (Q)
- ○ Total cost × Output (Q)
- ○ Marginal cost × Output (Q)

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Question # 4 of 10 (Start time: 11:36:00 AM, 04 July 2024)

Total Marks: 1

$$y = f(g(x))$$

$$\frac{dy}{dx} = \frac{d}{dx}\{f(g(x))\} = f'(g(x)).g'(x)$$

Given the above expression;

$$f'$$

represents:

Select the correct option

Reload Math Equations

- Inside function
- Derivative of inside function
- Outside function
- Derivative of outside function

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 03:54:55 PM, 05 July 2024)

Which of the following is true for the function:

$$Q = f(K(t), L(t))$$

?

Select the correct option

○	Capital and labor remain unhanged
○	Capital and labor do not depend on time
○	Capital and labor remain at their initial conditions
○	Capital and labor can improve over time

Click to Save Answe

Question # 7 of 10 (Start time: 03:53:17 PM, 05 July 2024)

Which of the following shows the differential of

$$y = f(x) = 4x^2 + 20$$

?

Select the correct option

○	$dy(8x) = dx$
○	$dy = (8x).dx$
○	$dy + dx = (8x)$
○	$dx = (8x).dy$

Click to